Homework #5

This is the homework for week 5. The due date is 11:59 p.m., October 20. Late submission will automatically result in "0" grade. Please include codes for the program part in your submission.

1 Computation Questions

(**Question1):** (20 points)

For a public opinion poll for a close presidential election, let p denote the proportion of voters who favor candidate A. How large a sample should be taken if we want the maximum error of the estimate of p to be equal to

(a) 0.03 with approximate 95% confidence?

(b) 0.02 with approximate 95% confidence?

(c) 0.03 with approximate 90% confidence?

(d) Since we do not know p in practice, please provide a valid upper bound for the number of samples in aforementioned problems which is independent of p. (Hint: you may use the fact that

$$\sqrt{p(1-p)} \leq \frac{p+1-p}{2} = \frac{1}{2}.$$

)

(**Question2):** (30 points)

Suppose $X_1, X_2, \cdots, X_n$ are i.i.d. Poisson random variables with parameter λ, i.e.,

$$Prob(X_1 = k) = \frac{\lambda^k}{k!} \exp(-\lambda) \text{ for } k \geq 0.$$

Recall the MLE for λ is $\frac{1}{n}\sum_{i=1}^{n} X_i$.

(a) Derive the Fisher information.

(b) Provide a valid estimator for the Fisher information. You do not need to prove the consistency of your estimator.

(c) Derive an $1-\alpha$ confidence interval for λ.

(d) Suppose we want the maximum error of the estimate of λ to equal 0.01 with approximate 95% confidence, determine the sample size. Your result can include λ.

(**Question3):** (30 points)

Suppose $X_1, \cdots, X_n$ are i.i.d. normal random variables following $N(0, a)$ distribution (mean 0, variance a).

(a) Derive the maximum likelihood estimator for a.

(b) Derive the Fisher information.

(c) Provide a valid estimator for the Fisher information. You do not need to prove the consistency of your estimator.

(d) Derive an $1-\alpha$ confidence interval for a.

2 Programming Questions

(20 points)

Implement the following experiment using Jupyter or R:

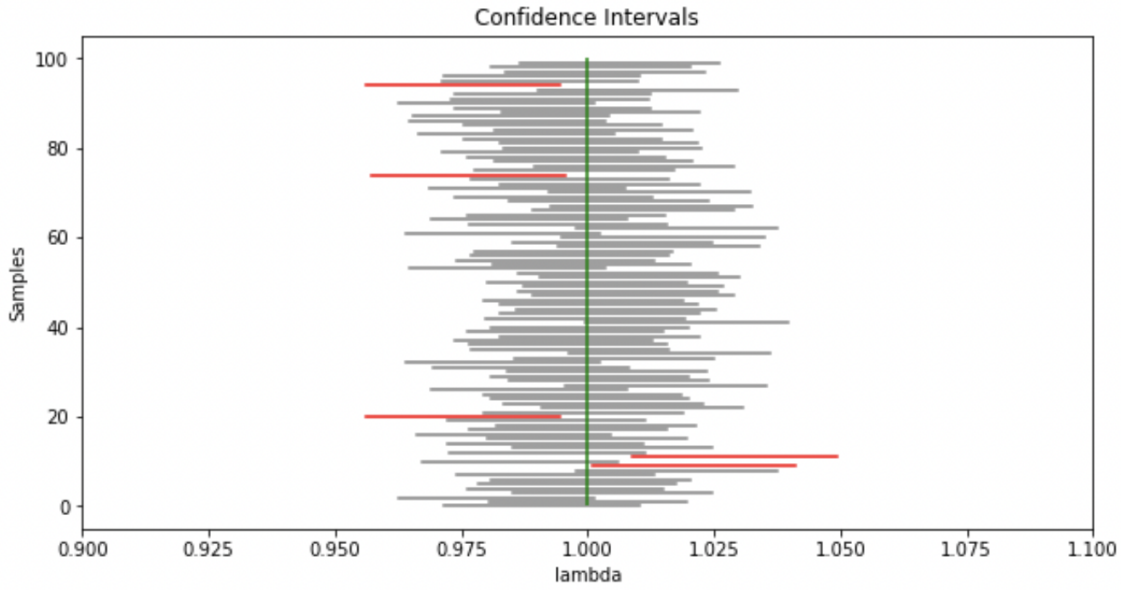
i. Generate $X_1, \cdots, X_n$ satisfying Bernoulli distribution with parameter $p = 0.3$ and $n = 40$.

ii. Plot the 95% confidence interval for p, i.e.,

$$\left[\widehat{p} - \frac{\sqrt{\widehat{p}(1-\widehat{p})}}{\sqrt{n}} \times z_{2.5\%},\ \widehat{p} - \frac{\sqrt{\widehat{p}(1-\widehat{p})}}{\sqrt{n}} \times z_{97.5\%}\right],$$

where z_α is the α upper quantile of a standard normal random variable.

iii. Repeat step i and ii for 100 times, see how many confidence intervals cover the real p. Your result should look like this



Please ignore the x–axis and y–axis scales of the figure, we just show what the result should look like.